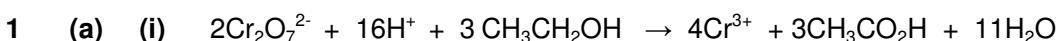


H1 Prelims 2013 Suggested Solutions

Paper 1

1	C	11	D	21	B
2	C	12	C	22	A
3	D	13	C	23	C
4	B	14	B	24	C
5	C	15	C	25	C
6	A	16	A	26	D
7	B	17	C	27	B
8	A	18	C	28	C
9	A	19	C	29	C
10	A	20	A	30	C

Paper 2 Section A

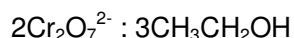


(ii) Orange to green

(iii) No. of moles of potassium dichromate

$$= 0.01715 \times 0.05$$

$$= 8.58 \times 10^{-4}$$



No. of moles of ethanol in 25 cm^3

$$= 8.58 \times 10^{-4} / 2 \times 3 = 1.287 \times 10^{-3}$$

(iv) No. of moles of ethanol in 250 cm^3

$$= 1.287 \times 10^{-3} \times 10 = 0.01287$$

No. of moles of ethanol in 25 cm^3 sample = 0.01287

No. of moles of ethanol in a can

$$= 0.01287 \times 330/25$$

$$= 0.1698$$

$$= 0.170 \text{ (3 s.f.)}$$

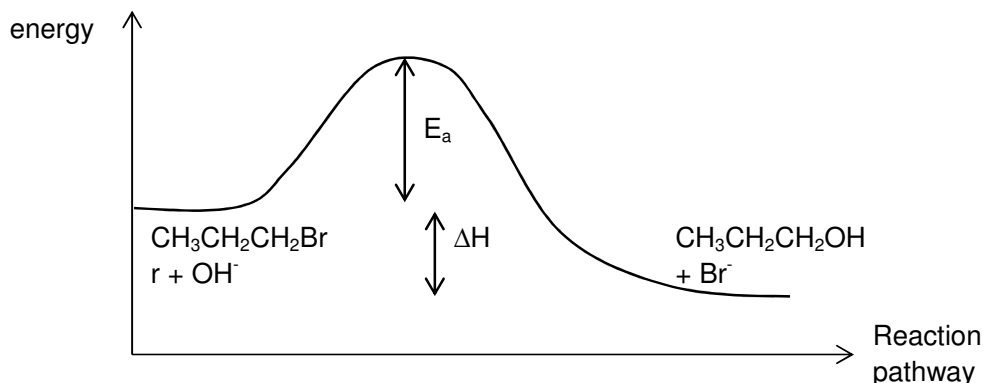
(v) $[\text{ethanol}] = 0.170 / 0.330 = 0.515\text{ mol dm}^{-3}$

- 2 (a)
- Butan-1-ol forms H-bonds as intermolecular forces
 - H-bonds stronger than the VDW between bromobutane molecules
- (b)
- $n(\text{butan-1-ol}) = (25 \times 0.81)/74 = 0.274\text{ mol}$
 - $n\text{NaBr} = 35 / 103 = 0.340\text{ mol}$
 - NaBr in excess
- (c)
- Theoretical Yield = $0.274 \times 137 = 37.5\text{g}$
 - Experimental Yield = $20 \times 1.35 = 27.0\text{g}$
 - % yield = $27 \times 100/37.5 = 72.0\%$

- (d) • loss of alcohol due to evaporation
 - formation of by-products
 - (e) • Reaction is exothermic
 - To prevent further oxidation of HBr or dehydration of alcohol
 - (f) • Substitution
 - (g) • Substitution involves breaking of strong covalent bonds in the reactants and hence heat energy is required.
 - (h) • Na metal
 - Brisk effervescence of hydrogen with butan-1-ol but not with 1-bromobutane.
- 3 (a) (i) Across the period, effective nuclear charge increases, hence the electrons in the outermost principal quantum shell experience stronger electrostatic forces of attraction to the nucleus. Atomic radius of Si is smaller than that of Al.
- (ii) Silicon is a giant covalent molecule, whereby the silicon atoms are held by strong covalent bonds. Phosphorus is a simple covalent non-polar molecule, whereby the molecules are held together by weak induced dipole-induced dipole interactions. Hence more energy is required to overcome the strong covalent bonds in Si, hence Si has a higher melting point.
- (b) (i) $\text{Si (s)} + 2\text{Cl}_2 \text{ (g)} \rightarrow \text{SiCl}_4 \text{ (l)}$
- (ii) $\text{SiCl}_4 \text{ (l)} + 4\text{H}_2\text{O (l)} \rightarrow \text{SiO}_2 \cdot 2\text{H}_2\text{O (s)} + 4\text{HCl (aq)}$
- Si has low-lying, vacant 3d-orbitals available to form dative covalent bonds with water molecules, hence SiCl_4 will hydrolyse completely in water.
- (c) (i) With the removal of each electron, the proton-to-electron ratio increases, hence remaining electrons are more strongly attracted to the nucleus, hence more energy is required to remove the subsequent electrons.
- (ii) 4th electron is removed from 3s-subshell, whereas 5th electron is removed from the inner 2p-subshell, which is of lower energy as compared to the previous. Hence, more energy will be required to remove the 5th electron.
- 4 (a) (i) Using expt 1 and 3, where $[\text{OH}^-]$ is kept constant, when [1-iodopropane] is tripled, rate is tripled. Hence first order wrt 1-iodopropane.
- Using expt 1,
- $$\text{Rate} = k[1\text{-iodopropane}][\text{OH}^-]$$
- $$1 = k (0.10) (0.20)^n$$
- Using expt 2,
- $$4 = k (0.20) (0.40)^n$$
- $$\frac{1}{4} = \frac{0.1(0.2)^n}{0.2(0.4)^n}$$
- $n = 1$ Hence it is first order wrt to OH^-

(ii) $\text{rate} = k [\text{1-iodopropane}][\text{OH}^-]$

(iii)



(b) (i) Test tube 1: 1-bromopropane

Test tube 2: 1-chloropropane

Test tube 3: 1-iodopropane

(ii) C-I bond is weakest and most readily broken hence the yellow ppt was obtained very quickly.

OR

C-Cl bond is the strongest and least readily broken hence the white ppt was obtained the slowest.

(c) (i) Heat evolved = $mc\Delta T = 300 \times 4.18 \times 10 = 12540 \text{ J}$

Amount of ethanol used = $0.4 / 60 = 6.67 \times 10^{-3}$

$\Delta H_c = -Q / n = -12540 / 6.67 \times 10^{-3} = 1881 \text{ kJ mol}^{-1}$

(ii) Heat loss to the surroundings.

(iii) Bond breaking = $2(350) + 7(410) + 360 + 460 + 9/2(496) = +6622 \text{ kJ mol}^{-1}$

Bond forming = $6(740) + 8(460) = -8120 \text{ kJ mol}^{-1}$

$\Delta H_c = +6622 - 8120 = -1498 \text{ kJ mol}^{-1} = -1500 \text{ kJ mol}^{-1}$

(iv) Heat of vaporization was not taken into account as the values of bond energies are meant for gaseous particles.

Paper 2 Section B

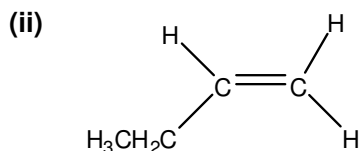
5 (a) (ii) Conc H_2SO_4 is a catalyst.

Conc H_2SO_4 is a dehydrating agent. It removes H_2O and POE shifts right generating more ester, thus increasing the yield.

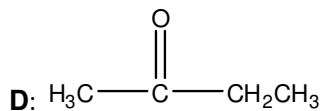
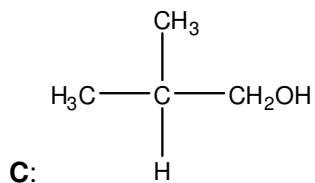
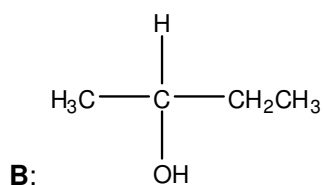
(iii)
$$K_c = \frac{[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3][\text{H}_2\text{O}]}{[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}][\text{CH}_3\text{CH}_2\text{COOH}]}$$

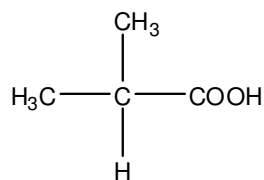
(iv) $K_c = (1.32)^2 / (0.48)^2 = 7.56 \text{ (dimensionless)}$

- (v) By LCP, increasing [butan-1-ol] causes equilibrium to shift to the right.
 K_c value remains unchanged.
- (vi) C_a : sp^2 , trigonal planar, 120°
 C_b : sp^3 , tetrahedral, 109.5°
- (b) (i) Step I: conc H_2SO_4 , $170^\circ C$
 Step II: conc H_2SO_4 $0^\circ C$ followed by boiling with H_2O

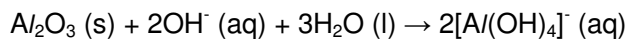
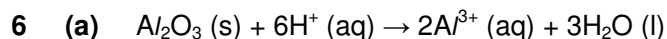


- (iii) No. There are 2 H atoms on the same sp^2 C.
- (c)
- | Evidence | Deduction |
|---|--|
| B and C has a formula $C_4H_{10}O$ which reacts with Na to give a gas | B and C are alcohol
Gas produced is H_2 |
| B and C reacts with manganate(VII) to give D and E | B and C is either a primary or secondary alcohol |
| D does not react with $NaHCO_3$ | D is not an acid |
| D reacts with 2,4-DNPH | D is a ketone |
| D reacts with aq alkaline I_2 | D has the structure $CH_3C=O$ |
| E reacts with $NaHCO_3$ | E is an acid |
| C reacts with hot conc H_2SO_4 | Elimination reaction |
| Alkene with hot $KMnO_4/H^+$ gives a neutral compound C_3H_6O | Oxidation reaction
Propanone is formed. |





E:



(ii) AlF_3 and Al_2O_3 are ionic compounds, while AlCl_3 and AlBr_3 are simple covalent molecules.

The magnitude of lattice energy is a measure of the strength of an ionic bond and

given as: $\text{LE} \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$.

Although ionic radius of F^- is smaller than ionic radius of O^{2-} , but O^{2-} carries a charge of -2, while F^- carries a charge of -1. Hence the ionic bond strength in Al_2O_3 is stronger and hence it has a higher melting point.

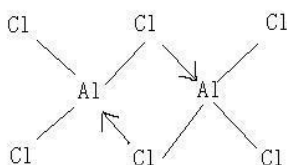
AlCl_3 has a smaller electron cloud size than that of AlBr_3 , hence is less polarisable, thus the induced dipole-induced dipole interactions are weaker in AlCl_3 and it has a lower melting point.



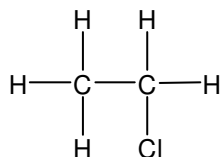
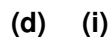
Shape of AlCl_3 : Trigonal planar

Bond angle: 120°

(ii)



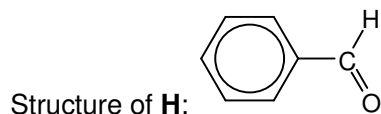
No. The chlorine atoms do not lie on the same plane.

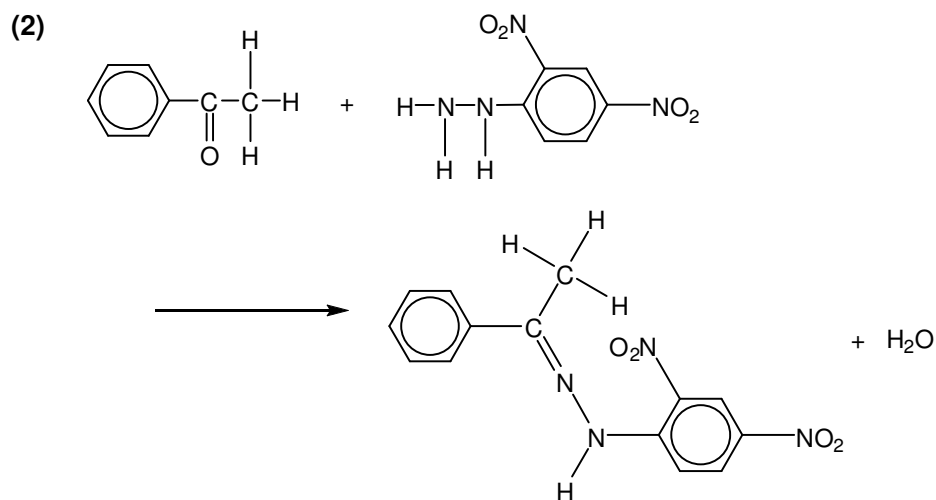
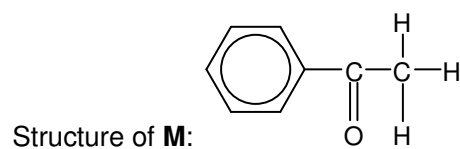
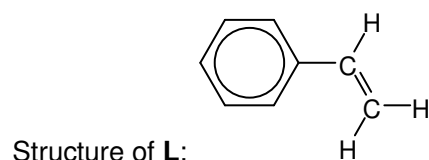
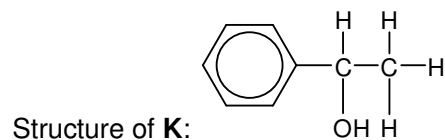
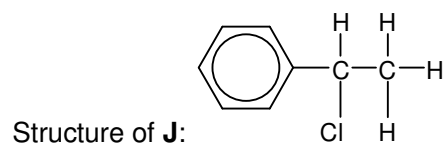


Structure of **F**:

Physical state of **F**: Liquid

(ii) (1)





- (3)
- $\text{NH}_3(\text{aq})$, $\text{AgNO}_3(\text{aq})$, warm (Tollen's Reagent)
 - Silver mirror will be formed for C but no silver mirror is formed for G
 - $\text{NaOH}(\text{aq})$, $\text{I}_2(\text{aq})$, warm (to 60°C)
 - Yellow crystals of CHI_3 are formed for G but not for C

- 7 (a) (i) • pK_a of 3-chloropropanoic acid would lower than that of propanoic acid
 • electron withdrawing chlorine helps to disperse the negative charge on the anion hence reducing its basicity
- (ii) • 2-chloropropanoic acid is stronger than 3-chloropropanoic acid
 • Proximity of electron withdrawing chlorine to the carboxylate increases the electron-withdrawing effect
- (b) (i) At half-neutralisation, the $pH = pK_a = 3.98$
- (ii) • $ClCH_2CH_2COOH + OH^- \rightarrow ClCH_2CH_2COO^- + H_2O$
 • $ClCH_2CH_2COO^- + H^+ \rightarrow ClCH_2CH_2COOH$
- (iii) • $[H^+] = (10^{-3.98} \times 0.4)^{1/2} = 6.47 \times 10^{-3}$
 • $pH = 2.19$
- (iv) • At neutralization, total volume = 90.0 cm^3
 • $[ClCH_2CH_2COO^-] = 0.02 / 0.09 = 0.222 \text{ mol dm}^{-3}$
 • $K_b (ClCH_2CH_2COO^-) = 10^{-14} / 10^{-3.98} = 9.55 \times 10^{-11} \text{ mol dm}^{-3}$
 • $[OH^-] = (9.55 \times 10^{-11} \times 0.222)^{1/2} = 4.60 \times 10^{-6} \text{ mol dm}^{-3}$
 • $pH = 8.66$
- (c) • Step 1 HCN, CN^- 10 – 20 °C
 • Step 2 PCl_5 , RT
 • Step 3 dilute H^+ , heat under reflux
- (d) • NaOH (aq), boil and acidify with $HNO_3(aq)$, followed by $AgNO_3(aq)$
 • White ppt with 3-chloropropanoic acid but not with propanoic acid